



AGENT ZERO

Production Readiness Engineering Checklist

PRODUCTION GRADE

ENTERPRISE ENGINEERING STANDARDS

To achieve robust, high-performance execution on hackathon day, the codebase must be hardened to production standards. This includes adding variables validation, runtime safety bounds, semantic deduplication, connection pooling, and complete fallback workflows across all 3 Git branches.

TEAM TASK CHECKLIST

Work on these tasks strictly inside your respective branches. Merge only through Pull Requests into `dev` when complete.

Person A — Orchestrator & Infrastructure Lead

feat/orchestrator

☐ **Zod-Based Environment Variables Validation**

Implement strict schema validation on startup to verify all key APIs (OpenAI, Gemini, Supabase, Twilio) exist, throwing clean fatal stack traces if missing.

☐ **Centralized JSON Persona Mappings**

Extract prompt strategies and roles out of javascript logic into a single `config/agents.json` file to allow rapid renaming and restructuring on demand.

☐ **Token Auditor & Cumulative Cost Tracker**

Add metadata trackers compiling exact tokens consumed by Llama router, Gemini, and Claude validator, writing cumulative costs to database logs.

☐ **Full-Stack Docker Compose Orchestration**

Write robust Dockerfiles and `docker-compose.yml` to bundle Next.js, Express, a local Redis session cache, and a PostgreSQL pgvector image.

☐ **Interactive LLM Runtime Swapper UI**

Add front-end control drop-downs letting developers dynamically override LLM endpoints (Gemini, Claude, Groq) per agent role on the fly.

☐ **Centralized Diagnostics `/health` Endpoint**

Write a detailed server health check showing network latency bounds, memory limits, Redis uptime, and Supabase connection latency.

⚙️ Person B — Integration & Tools Lead

feat/tools

☐ Strict Promise-Based Tool Timeouts

Wrap every dynamic tool call (e.g. Scraper, voice API, Skyvern) in a 15-second Promise.race timeout to prevent frozen agent processing loops.

☐ Full-Suite Mock Providers Switch

Implement `process.env.USE MOCKS` that intercepts Tavily search, WhatsApp, SMS, and Skyvern, routing requests to seed databases without calling live endpoints.

☐ Skyvern cloud Webhook Callback Receiver

Expose a POST endpoint in the Express server to handle asynchronous task updates from Skyvern and pipe status events to the frontend.

☐ Dynamic Model Context Protocol (MCP) Loader

Set up an MCP connection broker to map local system directory files and command line tools directly to the AI agent runtime.

☐ Redis-Backed Tool Invocation Rate Limiter

Add a token-bucket rate limiter limiting the maximum number of tool calls per execution to protect APIs from recursive infinite calls.

☐ Jest Automated E2E Tools Test Suite

Build robust testing specs validating that all 12 tools accept, execute, and respond with exact JSON schema outputs.



Person C — Database & Memory Lead

feat/memory

☐ Reciprocal Rank Fusion (RRF) Hybrid Search

Implement reciprocal rank scoring to merge semantic vector results with full-text keyword (BM25) search inside the Supabase client.

☐ Supabase Connection Pool Optimization

Configure pooling configurations using PgBouncer settings to prevent db exhaustion when under heavy benchmarking stress.

☐ Mem0 Fact Semantic Deduplication Loop

Write a database trigger or middleware to periodically run similarity cleanup on user personalization vectors, pruning duplicate facts.

☐ Recursive Folder Document Ingest Engine

Provide a robust console CLI tool to parse nested directory files (.pdf, .txt, .json), chunk semantically, and generate pgvector embeddings automatically.

☐ Clean Slate Database Reset Route

Implement `/api/memory/reset` Express endpoints to purge memory cache tables and rebuild vectors for quick sandbox resets.

☐ SQL Schema Migration Scripts

Write structured migrations (`supabase/migrations/`) documenting table setups, HNSW vector indices, and foreign key relations for clean setups.